

Quiz #1 Recap

February 18, 2026

TA: Alice

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Q9 Hypothesis statement

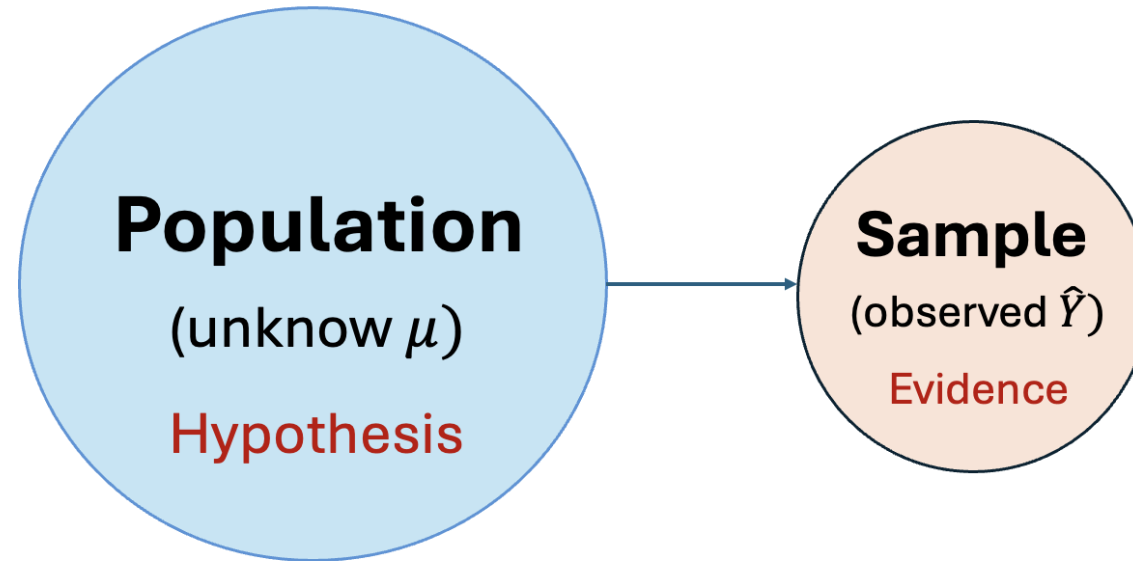
9. Please complete the sentence: A hypothesis is a statement that we make about... (Please select one.)

☐ ...a sample.

☒ ...a population.

Location: Week 4 –p1

Q9 Hypothesis statement



A **hypothesis** is always a *claim about a **population parameter*** (e.g., “the population mean equals 50”). We then collect **sample data** to evaluate whether the **evidence supports** or contradicts that claim.

Q10 Standard error of the mean

10. The "standard error of the mean" can be viewed as describing **sampling variability** and an estimate of the standard deviation of which one of the following? (Please select one.)

☐ it is the standard deviation of the variable of interest (e.g., the standard deviation of Y)

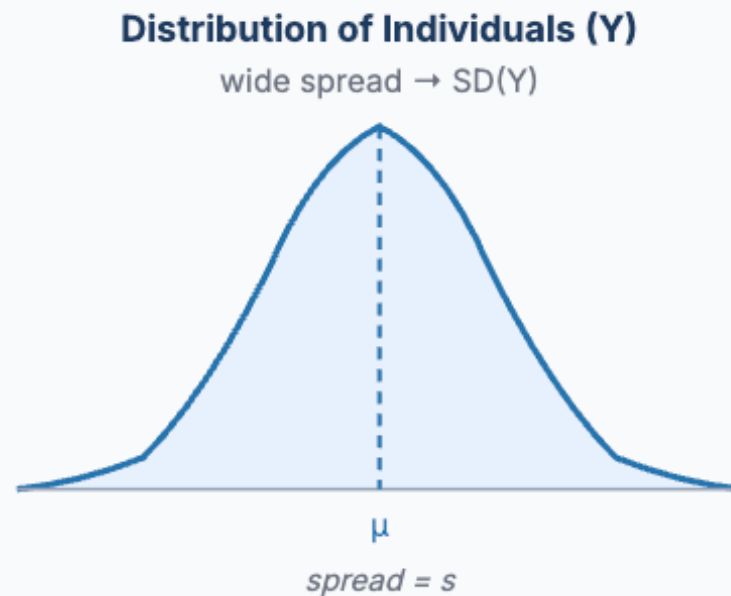
☒ it is the standard deviation of the variable of interest (e.g., the standard deviation of **sample means** (if one were to **draw many samples** and **obtain a sample mean** from each))

Location: Week 4 – p1

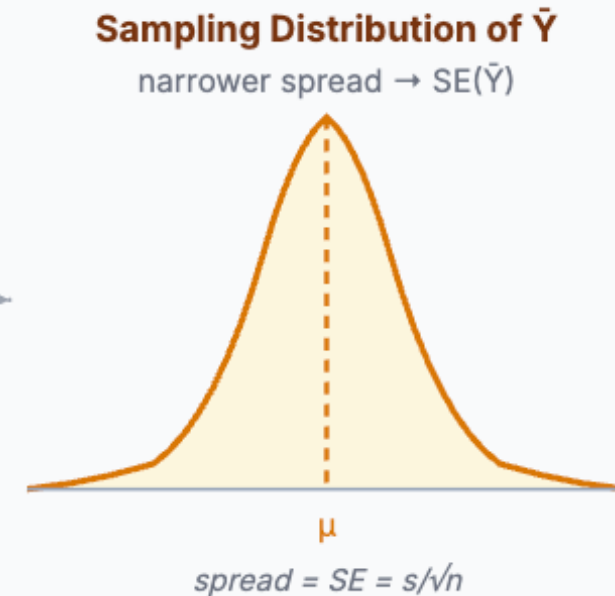
Q10 Standard error of the mean

SD of Y (often written s): How much **individual observations** scatter around the **sample mean**. This describes **person-to-person variability**.

SE of the mean: How much **sample means** would scatter if we repeated the study many times. Sampling variability, the **uncertainty** in our estimate.



$\div \sqrt{n}$
→



$$SE_{\bar{Y}} = \frac{\sigma_Y}{\sqrt{n}}$$

Q11 Units of a t-statistic

11. What are the units in which a t-statistic expresses a quantity (such as the difference between sample mean and hypothesized population mean)? (Please select one.)

☐ the units of the underlying variable

☒ standard error units (i.e., units of sampling variability)

Location: Week 4 – p1

Q11 Units of a t-statistic

The t-statistic re-expresses a raw difference in *standardized* terms. It answers: “How many **standard errors** away from the **hypothesized mean** is our **sample mean**?”

$$t_{obs} = \frac{\bar{Y} - \mu_0}{SE_{\bar{Y}}}$$

A **t value** of 2.5 means “the sample mean is 2.5 standard errors away from the hypothesized mean.”

Q12 p-value

12. What is a p-value? (Please select one.)

☐ it is the probability of that our hypothesis is correct

☐ it is the probability that our hypothesis is incorrect

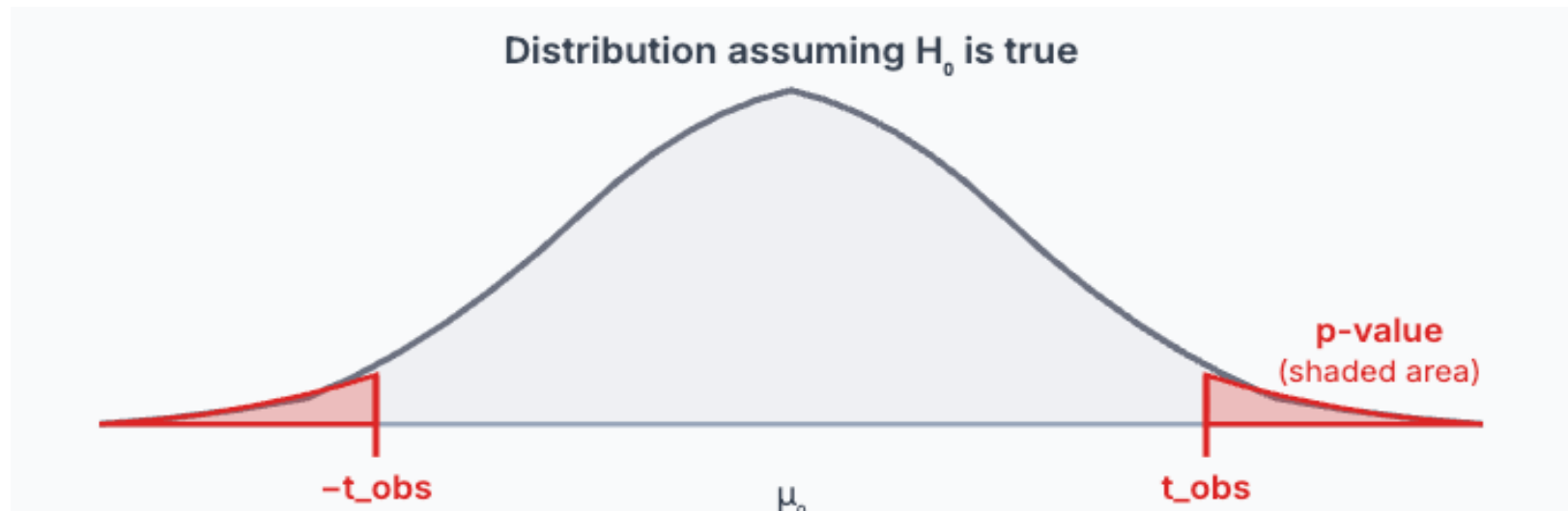
☒ neither of the above

Location: Week 4 – p2

Q12 p-value

A p-value is the **probability** of obtaining a test **statistic as extreme as (or more extreme than)** the **one observed**, assuming the **null hypothesis is true**.

It is *not* the probability that any hypothesis is correct or incorrect. The p-value tells us about the data (how surprising it is), not about the hypothesis itself.



Q13 Relationship between n and t_{crit}

13. What is the relationship between the sample size (n) for an analysis and the critical value t_{crit} (assuming we hold the alpha-level α constant)? (Please select one.)

- ☐ t_{crit} becomes smaller as the sample size increases
- ☐ t_{crit} becomes larger as the sample size increases

Location: Week 4 – p3 & p4

Q13 Relationship between n and t_{crit}

Intuition: With a **small sample**, we're **less sure** about our estimate of variability, so we need **stronger evidence** (a larger t_{obs}) to **reject H_0** . As **n grows**, we get a **more precise** estimate of variability, so the “bar” for significance lowers slightly.

Confidence Level	α	value of the t -statistic (“critical value,” t_{crit})			
		$df = \infty$	$df = 80$	$df = 20$	$df = 5$
80.0%	.20	1.282	1.298	1.325	1.476
90.0%	.10	1.645	1.664	1.725	2.015
95.0%	.05	1.960	1.990	2.086	2.571
99.0%	.01	2.576	2.639	2.845	4.032

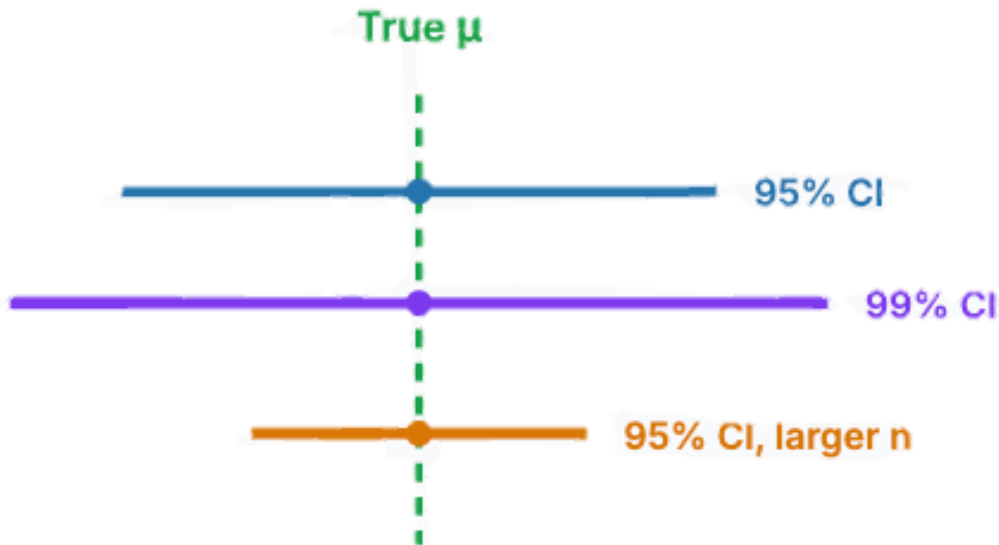
Q14 Increasing CI Coverage Rate

14. Which of the following would increase the proportion of samples in which the constructed Confidence Interval contains the population mean? (Please select all that apply.)

- ☐ select a lower confidence level
- ☒ select a higher confidence level
- ☐ increase the sample size
- ☐ decrease the sample size

Location: Week 4 – p5

Q14 Increasing CI Coverage Rate



The **confidence level**: the **proportion of samples** whose CI will **contain the true μ** . A 99% CI captures μ in 99% of samples; a 95% CI captures μ in 95% of samples. **Raising the confidence level** raises the **coverage rate**.

Increasing n makes the **CI narrower**, but it **does not change** the **proportion** of CIs that capture μ . $n = 30$ or $n = 3000$, a 95% CI will contain μ approximately 95% of the time.

Q15 Inference about population mean

15. Which of the following approaches would allow us to draw an inference concerning a hypothesis about the population mean?
(Please select all that apply.)

- ☐ comparison of a t-statistic (t_{obs}) to its critical value (t_{crit})
- ☐ comparison of the p-value to the alpha-level (α)
- ☐ comparison of the hypothesized population mean (μ_0) to the confidence interval around the sample mean
- ☐ comparison of the hypothesized population mean (μ_0) to the sample mean ($\bar{Y}$)

Location: Week 4 – p5

Q15 Inference about population mean

Approach 1

Compare t_{obs} to t_{crit}
Reject H_0 if $|t_{\text{obs}}| > t_{\text{crit}}$

Approach 2

Compare p-value to α
Reject H_0 if $p < \alpha$

Approach 3

Check if μ_0 is in the CI
Reject H_0 if μ_0 falls
outside CI

Not a valid approach

Compare μ_0 to $\bar{Y}$ directly
This ignores sampling variability.

Thank you for your attention!

